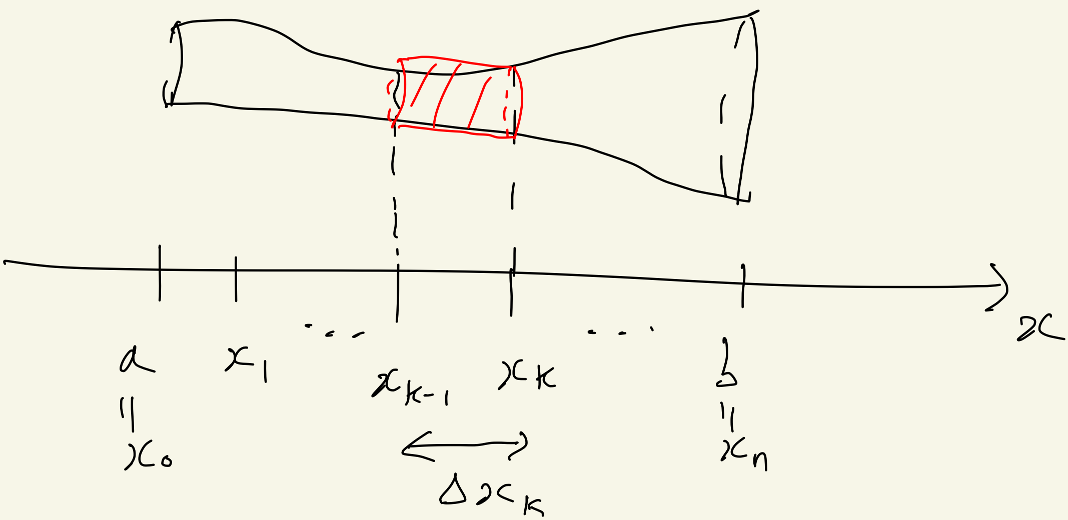
Volumes

Calculus A(1)
12/10

Homework 7 is online

Slicing



$$P = \{ x_0 < x_1 < \cdots < x_{n-1} < x_n \}$$

$x_0 = a$, $x_n = b$

Partition of $[a, b]$

$$\Delta x_k = x_k - x_{k-1}$$

$$V_k = A(x_k)\cdot \Delta x_k$$

Where: $A(x)$ = area of the cross-section of the solid at x

total volume

$$V \simeq \sum_{k=1}^{n} V_k = \sum_{k=1}^{n} A(x_k)\Delta x_k$$

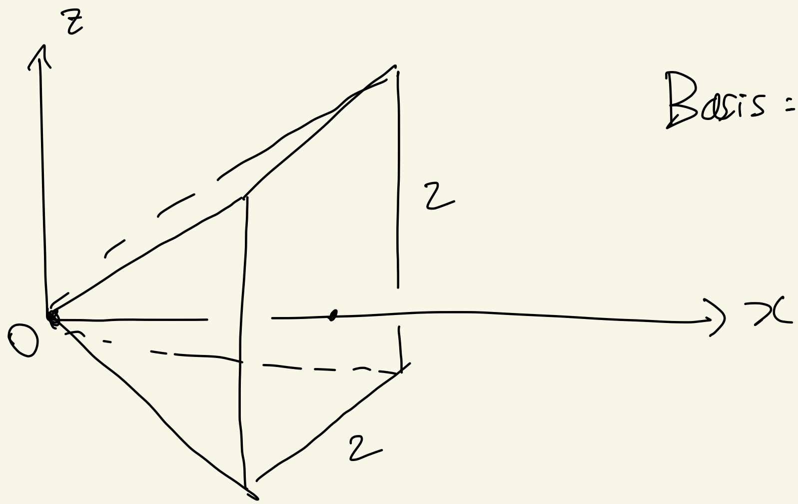
This is a Riemann sum for

$$x \longmapsto A(x).$$

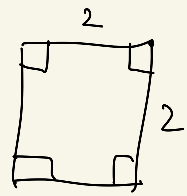
Def: Consider a solid whose cross-section at $x \in [a,b]$ has area $A(x)$. Assume $x \longmapsto A(x)$ is integrable on $[a,b]$ (e.g. $x \mapsto A(x)$ is C^0). Then we define the volume of the solid by

$$\boxed{V = \int_a^b A(x)\,dx}$$

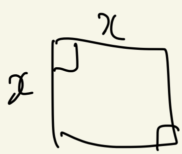
Ex : Volume of a pyramid.



Basis :



Let $x \in [0,2]$. The cross-section at x is a square $x \times x$



. So $A(x) = x^2$

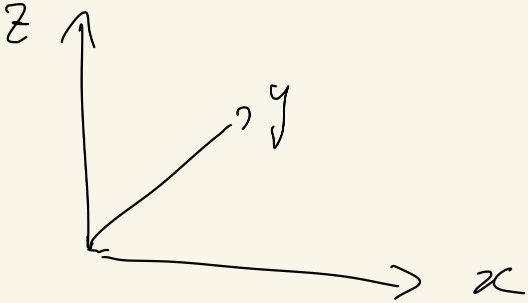
So $V = \int_0^2 A(x)\,dx$

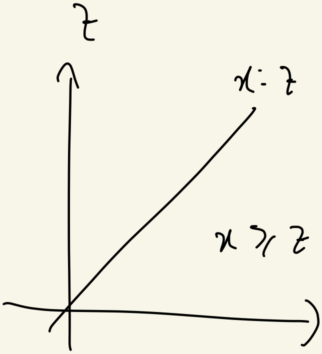
$$= \int_0^2 x^2\,dx = \left[\frac{1}{3}x^3\right]_0^2$$

$$= \frac{8}{3}.$$

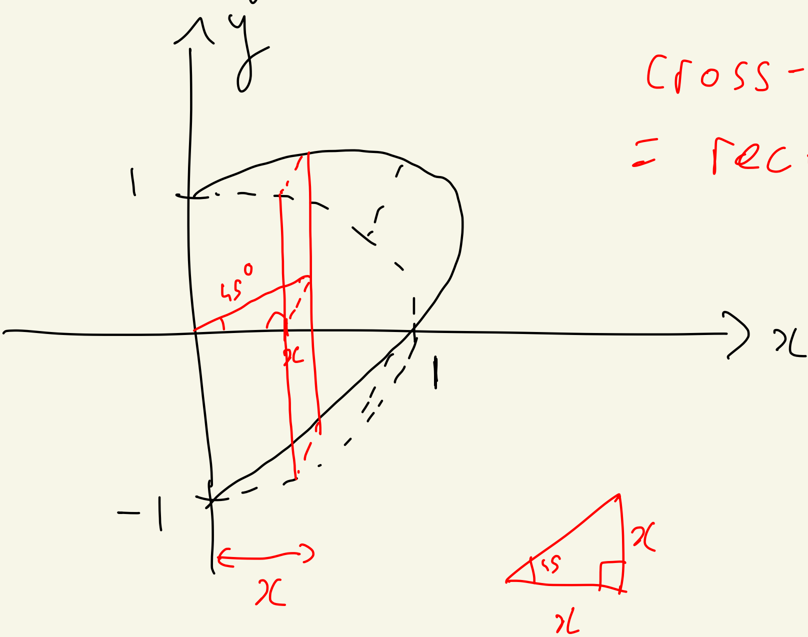
<u>Ex</u>: Start with a vertical cylinder whose base is the unit disc.

We cut this cylinder by the regions $x \geq z$ and $x \geq 0$

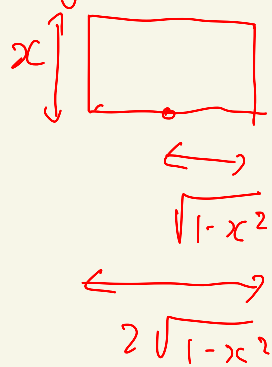




We get a "wedge":



Cross-section = rectangle



Cross-section at x is a rectangle whose dimensions are x and $2\sqrt{1-x^2}$

So $\quad A(x) = 2x\sqrt{1-x^2}$

$$V = \int_0^1 A(x)\,dx = \int_0^1 2x\sqrt{1-x^2}\,dx$$

$u = 1-x^2 \qquad du = -2x\,dx$

$$V = \int_1^0 -\sqrt{u}\,du = \int_0^1 \sqrt{u}\,du$$

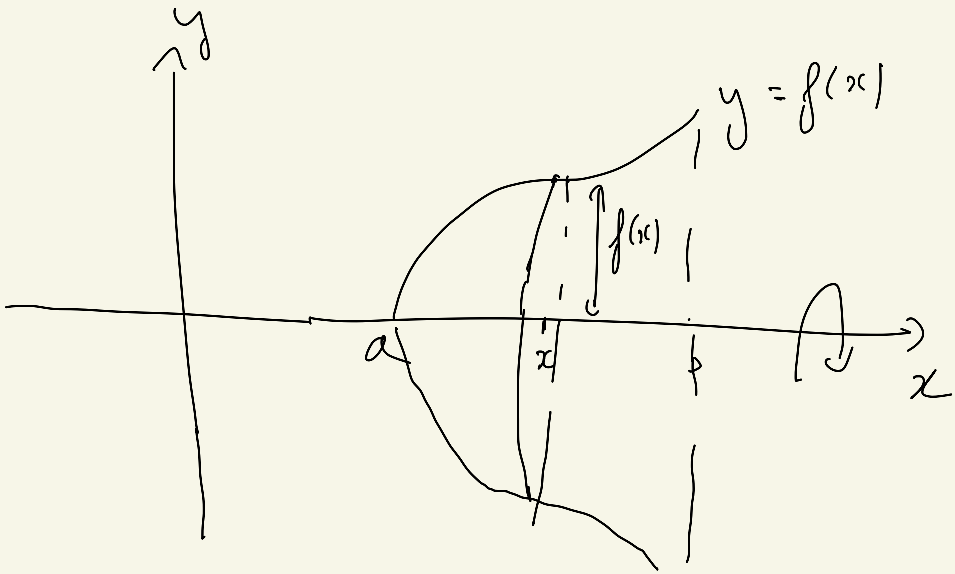
$$= \left[\frac{1}{1+\frac{1}{2}} u^{1+\frac{1}{2}}\right]_0^1$$

$$V = \frac{2}{3}$$

One can apply the slicing method to a special kind of solids: the ones obtained by rotating a plane region about an axis.

Such a solid is called a <u>solid of revolution</u>.

<u>Ex</u>:



The cross-section at x is a disc of radius $f(x)$

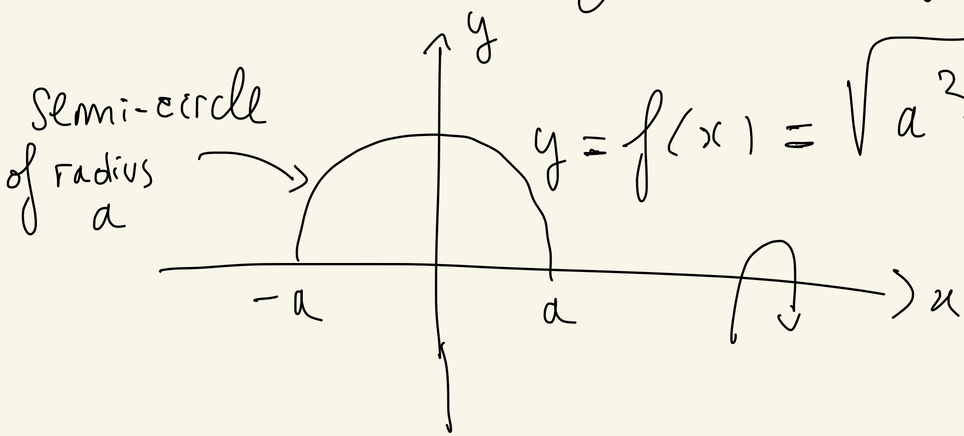
(Assume $f \geq 0$)

So $\underline{A(x)} = \pi f(x)^2$

So $$\boxed{V = \int_a^b \pi f(x)^2 \, dx}$$

is the volume of the region obtained by rotating $y = f(x)$ for $x \in [a,b]$ about the x-axis.

<u>Ex</u> (volume of a ball of radius a)



$y = f(x) = \sqrt{a^2 - x^2}$

$$V = \int_{-a}^{a} \pi f(x)^2 \, dx = \text{volume of ball of radius } a$$

$$V = \int_{-a}^{a} \pi (a^2 - x^2) \, dx$$

$$V = \int_{-a}^{a} \pi a^2 \, dx - \pi \int_{-a}^{a} x^2 \, dx$$

$$= \pi a^2 \cdot (a - (-a)) - \pi \left[\frac{1}{3} x^3 \right]_{-a}^{a}$$

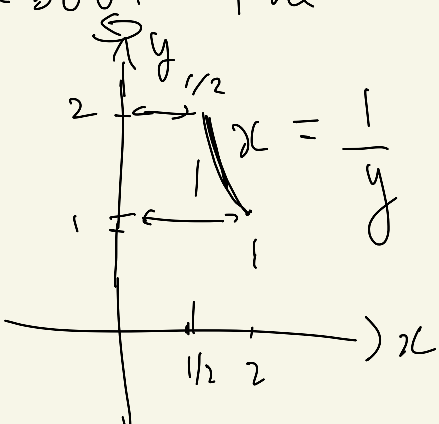
$$= 2\pi a^3 - \frac{2\pi}{3} a^3 = \frac{4\pi}{3} a^3.$$

So $\boxed{V = \frac{4\pi}{3} a^3}$

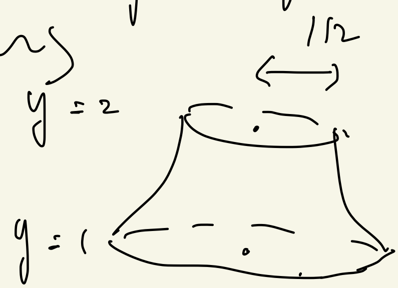
<u>Ex</u>: We rotate the graph

$x = g(y) = \frac{1}{y}$ for $y \in [1, 2]$

about the y-axis



rotating about y-axis



$$V = \int_1^2 \pi\, g(y)^2 \, dy$$

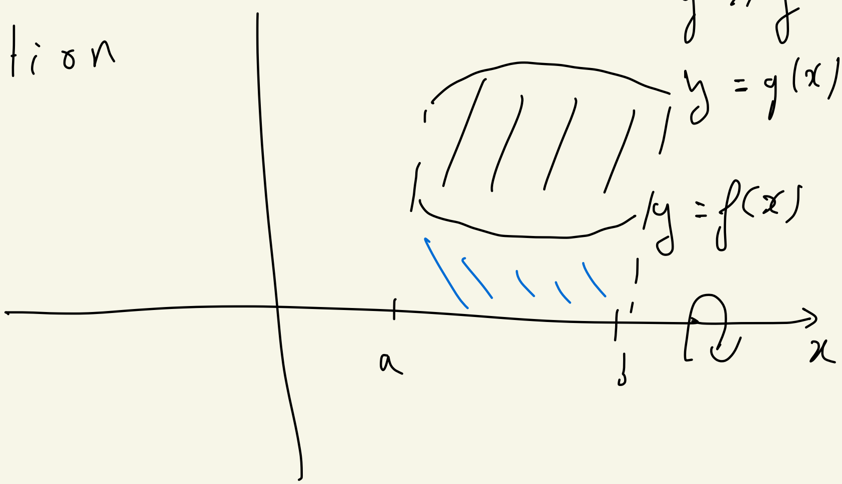
$$= \int_1^2 \pi \cdot \frac{1}{y^2} \, dy = \pi \left[-\frac{1}{y} \right]_1^2$$

$$= \frac{\pi}{2}$$

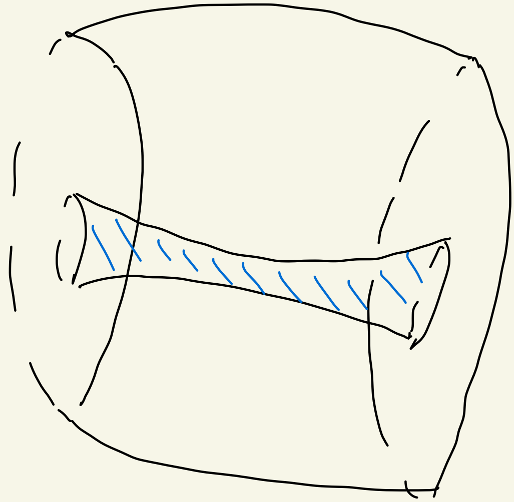
One can also rotate a region which does not border the axis of revolution

e.g

$g \geq f$



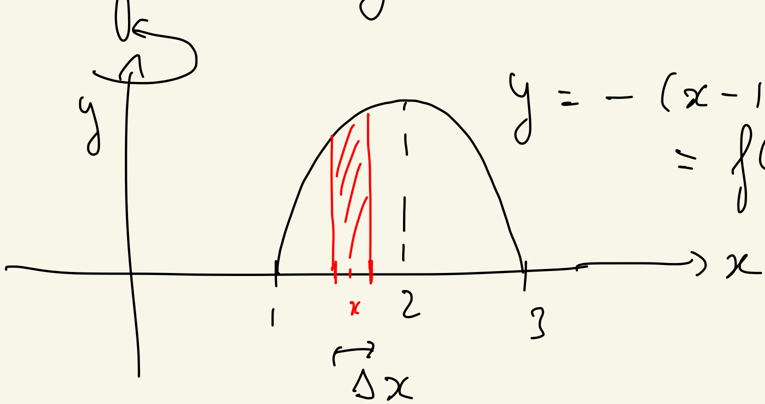
We get :



$$V = \int_a^b \pi \left(g(x)^2 - f(x)^2 \right) dx$$

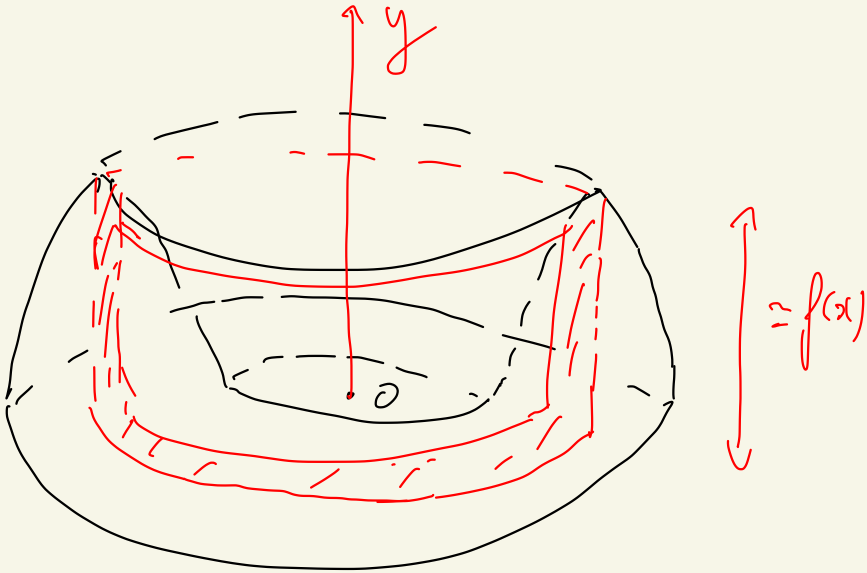
There is another way to slice the solid cylinders of increasing radii.

e.g

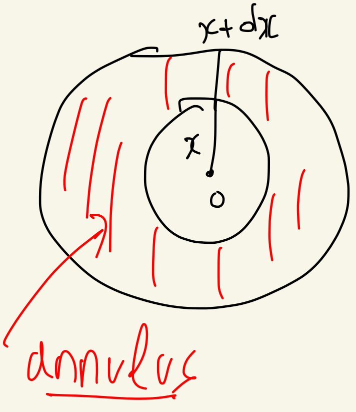


$y = -(x-1)(x-3)$
$= f(x)$

We get:



If Δx is "small", we have the following approximation for the red region: its volume is $\approx (2\pi x\, dx)\, f(x)$

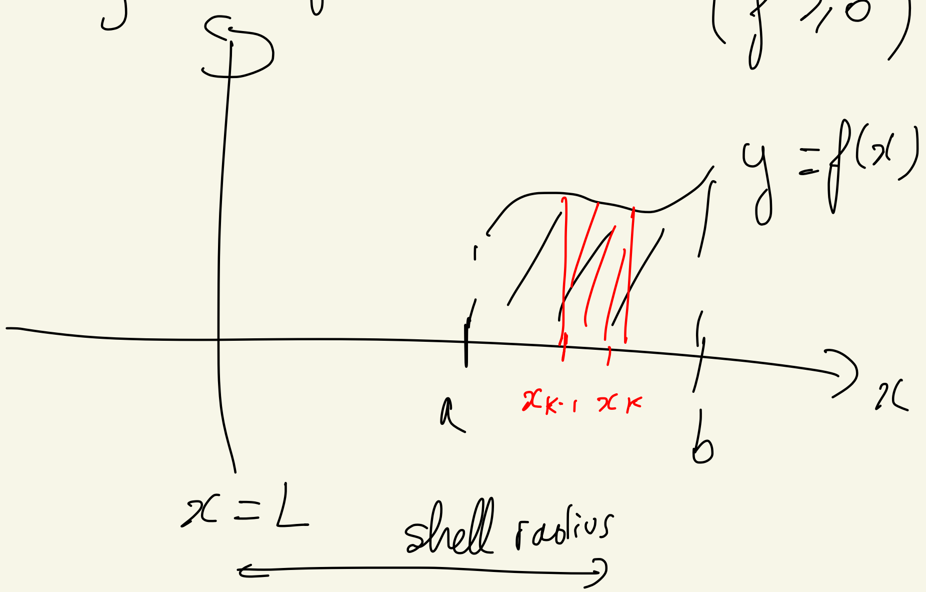


$$\text{area} = \pi(x+dx)^2 - \pi x^2$$
$$= \pi x^2 + 2\pi x\, dx + \pi (dx)^2 - \pi x^2$$
$$\simeq 2\pi x\, dx \quad \text{if } dx \text{ small}$$

Conclusion : $\boxed{V = \int_1^3 2\pi x f(x)\,dx}$

More generally, consider the following region:

$(f \geq 0)$



$L \in \mathbb{R}$

We rotate the region below $y = f(x)$ about the axis $x = L$

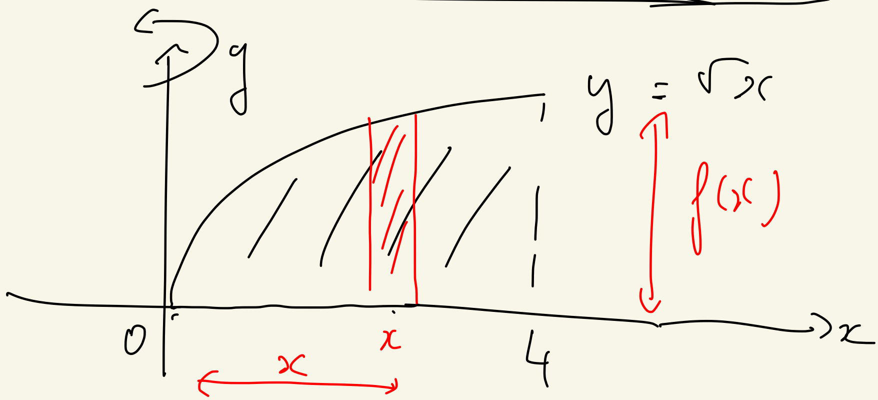
$(L \leq a)$

The volume we get is

$$V = \int_a^b 2\pi \underbrace{(\text{shell radius})}_{= \, x - L} \underbrace{(\text{shell height})}_{= \, f(x)} \, dx$$

$$\boxed{V = \int_a^b 2\pi (x - L) f(x) \, dx .}$$

Ex :



$$V = \int_0^4 2\pi x f(x) \, dx = \int_0^4 2\pi x^{3/2} \, dx$$

$$= 2\pi \left[\frac{1}{1+\frac{3}{2}} x^{1+3/2} \right]_0^4$$

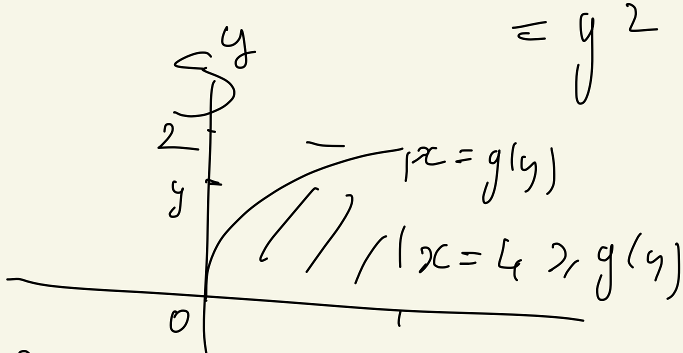
$$= 2\pi \cdot \frac{2}{5} \cdot 4^{5/2}$$

$$V = \frac{128\pi}{5}$$

<u>Remark</u>: One could also use the previous method for solids of revolutions

$$y = \sqrt{x} \iff x = g(y) = y^2$$

$(x, y \geq 0)$



$$V = \int_0^2 \pi (4^2 - g(y)^2)\, dy$$

$$V = 2 \cdot 4^2 \pi - \pi \int_0^2 g(y)^2 \, dy$$

$$= 32\pi - \pi \int_0^2 y^4 \, dy$$

$$= 32\pi - \pi \left[\frac{1}{5} y^5 \right]_0^2$$

$$= 32\pi - \frac{32\pi}{5} = \frac{128\pi}{5}$$

(§6.1 and 6.2)

(§6.3) **Lengths of plane curves**

We start with line segments.

$B = (x_B, y_B)$

$A = (x_A, y_A)$

length of AB

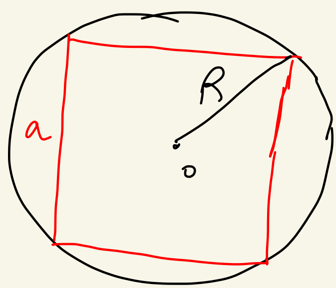
$$= |AB|$$

$$= \sqrt{\Delta x^2 + \Delta y^2}$$

$$= \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}$$

What about a general "curve"?

e.g. a circle



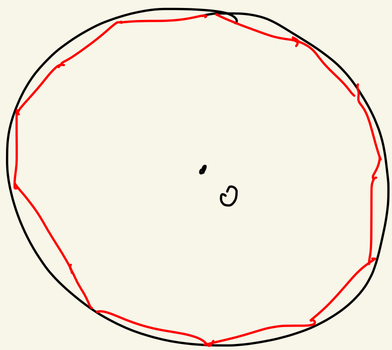
$a = \sqrt{2}\,R$

length of the square $= 4a = 4\sqrt{2}\,R$

This gives a first approximation of the length of the circle of radius R $(= 2\pi R)$

$$4\sqrt{2}\,R \simeq 2\pi R$$

Archimede used polygones to approximate π



More generally: let $\mathcal{C}$ be a parametric curve

$$x = f(t) \qquad t \text{ parameter}$$

$$y = g(t) \qquad t \in [a,b]$$

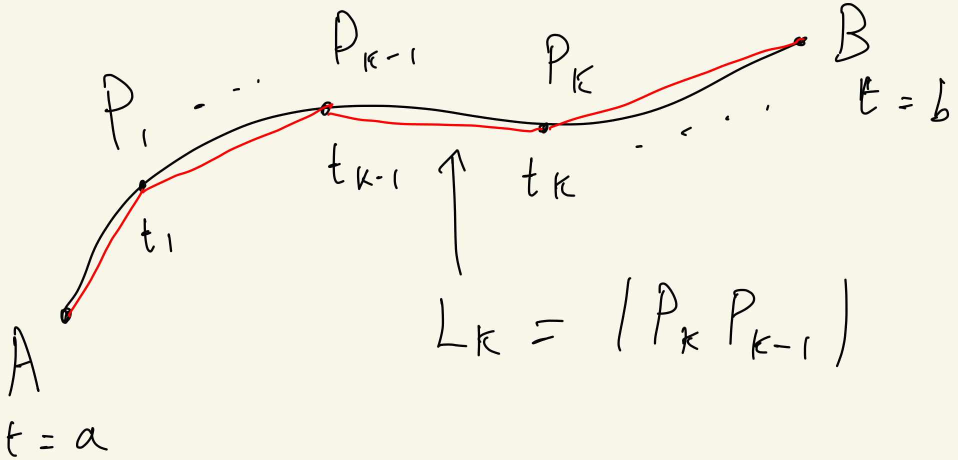
We shall assume f, g are C^1

(C^1: $\frac{df}{dt}$ and $\frac{dg}{dt}$ exist and are C^0)

We also assume that $\forall t \in [a,b]$ either $f'(t) \neq 0$ or $g'(t) \neq 0$

Such a curve C is called smooth

A smooth curve has a tangent line at every point.



$$P = \{ t_0 = a < t_1 < \dots < t_n = b \}$$

partition of $[a,b]$

Length L of the curve $\mathcal{C}$

is $L \approx \sum_{k=1}^{n} L_k$

where $L_k = \text{length } |P_k P_{k-1}|$

$= \sqrt{\Delta x_k^2 + \Delta y_k^2}$

$\Delta x_k = x_k - x_{k-1}$

$= f(t_k) - f(t_{k-1})$

and

$\Delta y_k = y_k - y_{k-1}$

$= g(t_k) - g(t_{k-1})$

MVT : $\exists\, t_k' \in [t_{k-1}, t_k]$

s.t $\frac{f(t_k) - f(t_{k-1})}{t_k - t_{k-1}} = f'(t_k')$

So $\Delta x_k = \Delta t_k \cdot f'(t_k')$

Similarly:

$$\Delta y_k = \Delta t_k \, g'(t_k'')$$

for some $t_k'' \in [t_{k-1}, t_k]$

So $L_k = \sqrt{\Delta x_k^2 + \Delta y_k^2}$

$$L_k = \left(\sqrt{f'(t_k')^2 + g'(t_k'')^2}\right) \cdot \Delta t_k$$

$\sum_{k=1}^{n} L_k$ is not a priori a Riemann sum because a priori $t_k' \neq t_k''$

$\rightsquigarrow$ one ~~desn't~~ doesn't have a single point in $[t_{k-1}, t_k]$.

By our assumption that f', g' are C^0, one can show that

$$\sqrt{f'(t_k')^2 + g'(t_k'')^2}$$

$$\simeq \sqrt{f'(c_k)^2 + g'(c_k)^2} = h(c_k)$$

for any $c_k \in [t_{k-1}, t_k]$

One can prove that

$$\lim_{n \to +\infty} \sum_{k=1}^{n} L_k = \int_a^b \sqrt{f'(t)^2 + g'(t)^2}\, dt$$

Thus, we are justified to

make the following definition:

Def: Let $\mathcal{C}$ be a smooth parametric curve $\begin{cases} x = f(t) \\ y = g(t) \end{cases}$ $t \in [a, b]$

Assume that $\mathcal{C}$ is traversed exactly once (no loops)

Then we define the length of $\mathcal{C}$ by

$$\boxed{L = \int_a^b \sqrt{f'(t)^2 + g'(t)^2}\, dt}$$

Remarks: ① This integral is well-defined because we have assumed f' and g' are C^0; so $\sqrt{(f')^2+(g')^2}$ is C^0.

② Other formulation

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

Not rigorous:

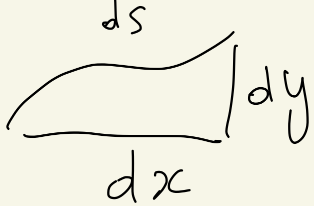
$$\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \sqrt{\left(\frac{dx}{dt}\right)^2 (dt)^2 + \left(\frac{dy}{dt}\right)^2 dt^2}$$

$$= \sqrt{dx^2 + dy^2}$$

if we write $ds = \sqrt{dx^2 + dy^2}$

We get simply: $L = \int_{\mathcal{C}} ds$



ds = arc length differential

(2) One can show that L does not depend on the smooth parametrization of $\mathcal{C}$ (cf. HW8).

(3) Case where $\mathcal{C}$ = graph $y = g(x)$
$x \in [a,b]$

$$\begin{cases} x = t = f(t) \\ y = g(x) = g(t) \end{cases}$$

$$\Rightarrow \quad L = \int_a^b \sqrt{1 + g'(x)^2}\, dx$$

Similarly, if $x = f(y)$

$$\begin{cases} x = f(y) = f(t) \\ y = t = g(t) \end{cases} \qquad y = t \in [c, d]$$

$$L = \int_c^d \sqrt{f'(y)^2 + 1}\, dy$$

<u>Examples</u> :

(1) Circle of radius r and center O

$$\begin{cases} x = r\cos(\theta) & \theta \in [0, 2\pi] \\ y = r\sin(\theta) \end{cases}$$

$$L \overset{def}{:=} \int_0^{2\pi} \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta$$

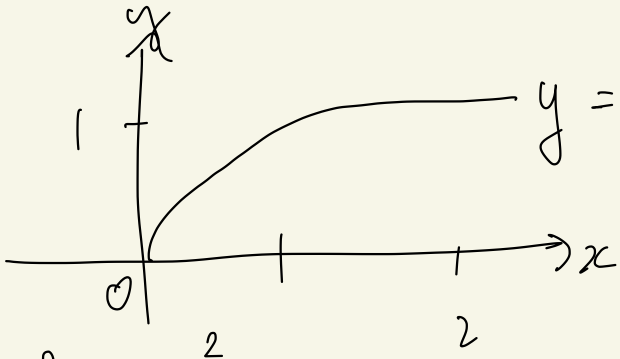
$$= \int_0^{2\pi} \sqrt{r^2 \sin(\theta)^2 + r^2\cos(\theta)^2}\, d\theta$$

$$= r \int_0^{2\pi} \sqrt{\underbrace{\sin(\theta)^2 + \cos(\theta)^2}_{= 1}}\, d\theta$$

$$= r \int_0^{2\pi} d\theta = 2\pi r$$

(it is how we defined π actually)

(2)



$y = \left(\frac{x}{2}\right)^{2/3}$

$= f(x)$

$$L \overset{?}{=} \int_0^2 \sqrt{f'(x)^2 + 1}\, dx$$

Problem: $f'(0)$ does not exist!

rewrite: $y = f(x)$ $\quad y^{3/2} = \frac{x}{2}$

as $x = g(y)$ where

$g(y) = 2y^{3/2}$ diff. on $[0,1]$

We have:

$$L = \int_0^1 \sqrt{g'(y)^2 + 1}\, dy$$

$$g'(y) = 2 \cdot \frac{3}{2} y^{3/2 - 1}$$

$$= 3 y^{1/2}$$

$$L = \int_0^1 \sqrt{9y + 1} \, dy.$$

$$u = 9y + 1$$

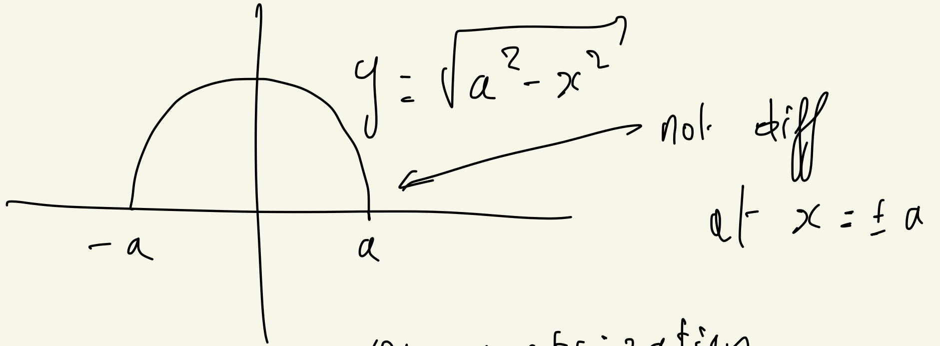
$$L = \frac{1}{9} \int_1^{10} \sqrt{u} \, du$$

$$= \frac{1}{9} \left[\frac{1}{1 + \frac{1}{2}} u^{1 + \frac{1}{2}} \right]_1^{10}$$

$$= \ldots$$

$$\sqrt{9y+1} \overset{\text{antiderivative}}{\rightsquigarrow} \frac{1}{9} \cdot \frac{1}{\frac{1}{2} + 1} (9y+1)^{3/2}$$

(3)



$L = ?$

parametrization

$x = a\cos(\theta)$

$\theta \in [-\pi, \pi]$ $\quad y = a\sin(\theta)$

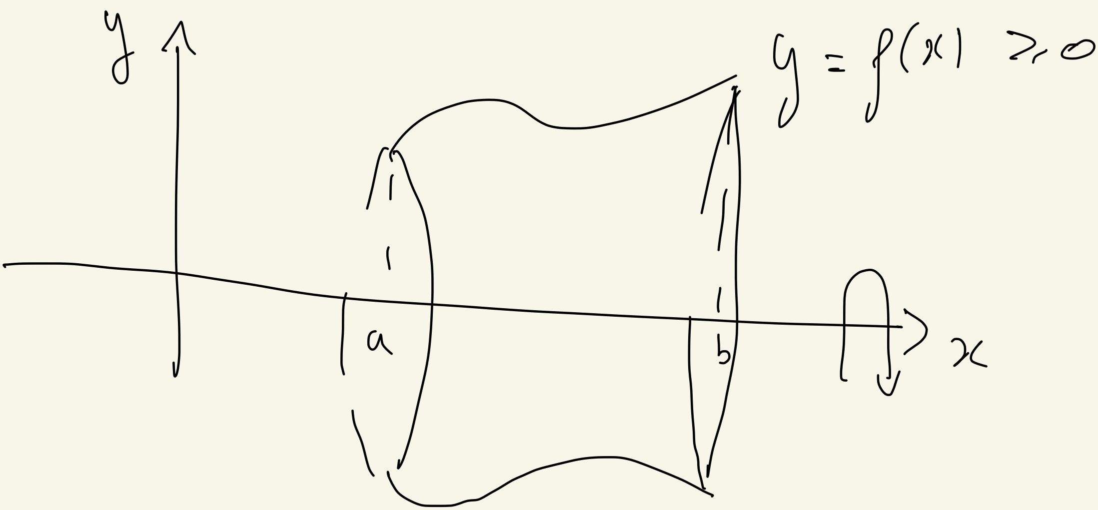
6.1 $\longrightarrow$ 6.3

6.5 areas

Areas of surfaces of revolutions

One wants to compute areas of

Surfaces obtained by rotating a curve about an axis.

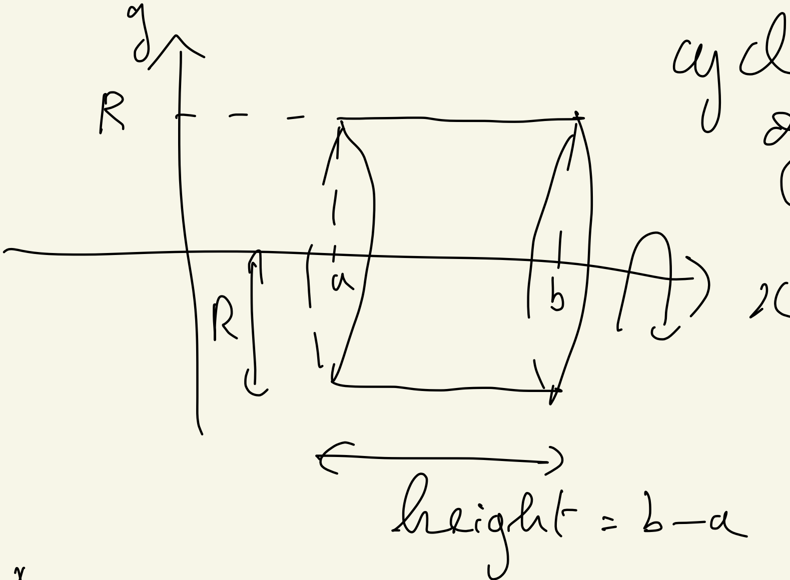


A = area of that surface.

$A = ?$

Let us first consider the simplest cases

(1) Case $f(x) = \text{constant} = R > 0$



cylinder of radius R and height $= b-a$

$$A = \binom{\text{circumference}}{\text{base}} \cdot (\text{height})$$

$$A = 2\pi R \cdot (b-a)$$

<u>Explanation</u>: One makes a horizontal "cut" and then "roll out" the cylinder.

One gets a rectangle:

$$A = (b-a)\cdot 2\pi R.$$